Construction Chemicals

SUBMITTAL DOCUMENT

Back-fill

Closed-Cell Polyethylene Foam Backer Rod for Joints



Depth Control
Correct sealant profile

Bond Breaker
No three-sided adhesion

Closed Cell
No water absorption

Recovery
> 90 %

Project	
Main Contractor	
Consultant	
Submittal No. / Rev.	00
Date	Nov 2026

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METHOD STATEMENT

Back-fill — Closed-Cell Polyethylene Foam Backer Rod for Joints

Project Name		**Document No.**	OMX-MS-BACKFILL
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
Product	Back-fill	**Status**	For Approval

1. Purpose

This method statement defines the procedure, sequence, resources, quality checks and safety measures for the supply and application of Back-fill on site. It ensures that the work is carried out safely, in line with the approved drawings and project specification, and achieves the performance stated in the O MIX Technical Data Sheet.

2. Scope of Work

This method statement covers the selection and installation of Back-fill closed-cell polyethylene backer rod in expansion and control joints before the application of joint sealants.

- Inspection and cleaning of joints.
- Selection of rod diameter and sealant depth.
- Installation of the backer rod.
- Inspection before sealing.
- Coordination with sealant application.

3. Definitions

Term	Meaning
Backer rod	Foam rod placed in a joint to control sealant depth.
Bond breaker	Material that prevents the sealant bonding to the joint bottom.
Three-sided adhesion	Sealant bonded to both faces and the bottom; causes failure under movement.
Width : depth ratio	Ratio of sealant width to its depth.

4. Roles & Responsibilities

Role	Responsibility
Project Manager	Overall responsibility for planning, resources, approvals and coordination with the consultant and main contractor.
Site Engineer	Supervises the work daily; confirms substrate readiness, climate conditions, mixing, consumption and sequence; keeps daily records.

Role	Responsibility
QA/QC Inspector	Carries out inspections and tests at each stage, records results, raises inspection requests at hold points and signs the checklist.
HSE Officer	Conducts toolbox talks, checks PPE, ventilation, access and safe storage; stops work if unsafe conditions arise.
Foreman / Applicators	Carry out the work as per this method statement and report any deviation immediately.
Storekeeper	Receives, stores and issues materials in original packaging under the stated storage conditions; controls batch numbers.

5. Materials

Material	Description
Back-fill	Closed-cell PE backer rod, diameters to suit joint widths
Joint sealant	Mixseal PU or as specified

5.1 Key product data

Property	Typical Value
Material	Closed-cell polyethylene foam
Density	35 – 40 kg/m³
Colour	Blue / black
Tensile strength	70 kPa
Elongation at break	116 %
Compression recovery	> 90 %
Flexibility	down to -40 °C
Thermal conductivity	0.043 W/m·K

6. Tools & Equipment

- Blunt insertion roller or wooden tool
- Knife
- Tape measure
- Depth gauge
- Wire brush
- Blower
- Thermo-hygrometer and surface thermometer (dew point check)
- Moisture meter
- Measuring buckets and scale
- Masking tape and protective sheeting
- Lighting and extension cables

6.1 Personal protective equipment

- Gloves
- Safety glasses
- Safety shoes

7. Storage & Handling

- Store in a dry, shaded place away from direct sunlight, heat sources and sharp objects.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Store rolls and cartons flat, off the ground, protected from direct sunlight, heat, oil and mechanical damage.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

8. Pre-Start Requirements

- Material submittal, this method statement and the inspection plan approved by the consultant.
- Work area handed over clean, free of other trades and with safe access.
- Adequate lighting, power supply and ventilation available.
- Climate conditions checked and recorded before work starts.
- A trial area prepared and approved where required by the project.
- Toolbox talk held with the work team covering this method statement and the safety measures.

9. Surface Preparation

1. Measure joint widths and confirm the movement design.
2. Clean joints of dust, laitance, old sealant, oil and loose material.
3. Ensure joint faces are dry and sound; repair damaged arrises.
4. Obtain QA/QC approval of cleaned joints before installing the rod.

10. Selection of Rod Diameter

1. Check the diameter and condition of the delivered rolls.
2. Select a rod diameter about 25 – 30 % larger than the joint width (compressed to about 70 % of its diameter).
3. Under water pressure, select a diameter that gives about 40 % compression.
4. Set the sealant depth: width 6 – 12 mm → depth = width; 12 – 24 mm → 12 mm; > 24 mm → half the width.

11. Installation

1. Unroll the rod without stretching.
2. Press the rod into the joint with a blunt roller or wooden tool to the set depth.
3. Keep a uniform depth along the joint; check with a depth gauge.

4. Butt-join ends tightly without overlapping.
5. Apply the joint sealant over the rod as soon as possible.
6. Record joint widths, rod diameters and depths.

12. Protection

- Protect installed rods from water, dirt and damage until sealed.
- Replace punctured or stretched rods before sealing.

13. Quality Control & Inspection

Check	Stage	Acceptance Criteria	By
Joints	Before installation	Clean, dry, sound	QA/QC
Rod diameter	Before installation	Compressed to ~70 % in joint	Site Engineer
Depth	After installation	Uniform, as per width : depth rule	QA/QC
Continuity	After installation	No gaps; ends butt-joined	QA/QC

13.1 Inspection and test plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Diameter, condition	Each delivery	Material inspection report
Joint cleaning	Clean, dry, sound	Each joint	Inspection request
Rod installation	Diameter, depth, continuity	Each joint	Inspection request (hold point)
Sealing	Sealant profile	Each joint	Daily report

14. Defects & Remedies

Defect	Likely Cause	Remedy
Rod too loose	Diameter too small	Replace with larger diameter
Uneven sealant depth	Rod inserted unevenly	Re-set rod to uniform depth
Bubbles in sealant	Punctured rod releasing gas	Use blunt tools only; replace punctured rod
Sealant three-sided adhesion	Rod missing or gaps	Install continuous rod

15. Health, Safety & Environment

Hazard	Control Measures
Cuts from knives	Cut-resistant gloves
Work at height	Approved access, harness where required

Hazard	Control Measures
Fire (combustible foam)	Keep away from heat and flames
Slips and trips	Housekeeping

- Wear gloves when handling and cutting.
- Keep away from heat sources and open flames (combustible material).
- Follow site safety rules for work at height.
- Eyes: rinse with clean water if particles enter the eye.
- Cuts: clean and dress the wound.

15.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

16. Cleaning

- No cleaning required; collect off-cuts in waste bins.

17. Limitations

- Do not puncture, twist or stretch the rod.
- Do not install in wet joints.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Joint widths measured	☐ Yes ☐ No	
Joints cleaned and dry	☐ Yes ☐ No	
Rod diameter selected	☐ Yes ☐ No	
Rod inserted with blunt tool	☐ Yes ☐ No	
Uniform depth verified	☐ Yes ☐ No	
Ends butt-joined	☐ Yes ☐ No	
No punctures	☐ Yes ☐ No	
Sealant applied promptly	☐ Yes ☐ No	

Inspected by	Signature	Date

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Technical Data Sheet

Back-fill — Submittal Document

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Product Datasheet
Edition: 01 | Date: Nov 2026
Jointing and Tiling

Back-fill

Closed-Cell Polyethylene
Foam Backer Rod for Joints



Product Description

Back-fill is a flexible, closed-cell polyethylene foam backer rod installed behind sealants in expansion and control joints. It controls the sealant depth, gives the correct sealant profile and acts as a bond breaker to prevent three-sided adhesion. It does not absorb water and is compatible with most joint sealants.

Uses

- Expansion and control joints in concrete and masonry.
- Joints in façades, curtain walls and precast panels.
- Floor joints before sealing.
- Joints around windows and doors.
- Backing for Mixseal PU and other O MIX sealants.

Advantages

- Controls sealant depth and profile.
- Natural bond breaker; prevents three-sided adhesion.
- Closed cells: no water absorption.
- Flexible; recovers more than 90 %.
- Remains flexible down to -40 °C.

Application

1. **Select Diameter**
   Choose a rod diameter so it is compressed to about 70 % of its diameter in the joint (40 % compression under water pressure).
2. **Insert**
   Press the rod into the joint with a blunt tool to a uniform depth; do not puncture or stretch it.
3. **Seal**
   Apply the sealant over the rod at the specified width : depth ratio.

Consumption

1 m per 1 m of joint
rod diameter selected to joint width
Do not stretch the rod during installation.

Packaging

Rolls in various diameters, or profiles up to 30 mm thick × 2 m long.

Storage & Shelf Life

Store in a dry, shaded place away from direct sunlight, heat sources and sharp objects.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Material	Closed-cell polyethylene foam	—
Density	35 – 40 kg/m³	—
Colour	Blue / black	—
Tensile strength	70 kPa	—
Elongation at break	116 %	—
Compression recovery	> 90 %	—
Flexibility	down to -40 °C	—
Thermal conductivity	0.043 W/m·K	—
Water absorption	Negligible (closed cell)	—

Consumption

Joint width	Sealant depth
6 – 12 mm	Equal to joint width
12 – 24 mm	12 mm
> 24 mm	Half of joint width

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Joint faces clean, dry and sound before installing the rod.
- Rod compressed to about 70 % of its diameter (40 % under water pressure).
- Install at a uniform depth along the joint.
- Do not puncture, twist or stretch the rod.

Application Notes

- Use a blunt roller or wooden tool to insert the rod; never a sharp tool.
- Apply primer to joint faces only, not to the rod, where the sealant requires a primer.
- Apply the sealant soon after installing the rod.
- Butt-join rod ends; do not overlap.

Health & Safety

- Wear gloves when handling and cutting.
- Keep away from heat sources and open flames (combustible material).
- Follow site safety rules for work at height.

First Aid & Cleaning

- Eyes: rinse with clean water if particles enter the eye.
- Cuts: clean and dress the wound.
- Burns from contact with melted foam: cool with water and seek medical advice.
- No cleaning required; collect off-cuts in waste bins.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.

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Test Reports

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To be attached: laboratory test reports supporting the values in the Technical Data Sheet.

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Approvals & Certificates

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To be attached: approvals and certificates, where available.